

Unpil Baek

RESEARCH ENGINEER – SCIENTIFIC COMPUTING & QUANTUM ALGORITHMS

Physics PhD-trained researcher with a background in quantum algorithms for electronic structure simulation and large-scale numerical optimization. Skilled in building robust research software and computational pipelines across quantum and classical workflows, with strengths in numerical linear algebra, variational methods, and performance-oriented implementation.

EDUCATION

University of California, Berkeley

Berkeley, CA

*M.A. & Ph.D. in Physics, GPA: 4.0/4.0**Aug 2017 – Aug 2023*

- **Concentrations:** Quantum algorithms, electronic structure simulations, tensor network methods
- **Dissertation:** “Exploring electronic structure with near-term quantum computers”
- **Coursework:** Quantum Computing, Electronic Structure Theory, Bayesian Data Analysis and Machine Learning for Physical Science

University of California, Berkeley

Berkeley, CA

*B.A. in Physics & Applied Math, GPA: 4.0/4.0**Aug 2013 – May 2017*

- **Thesis:** “Einstein-Cartan action under Gotay-Nester algorithm”
- **Awards:** Graduated *summa cum laude* in both; Departmental Citation in Physics

SKILLS

Programming: Python, C++, Julia, MATLAB, Mathematica**Libraries:** NumPy, Pandas, ITensor, OpenFermion, TensorFlow, PyTorch**Utilities:** Git (GitHub, GitLab), JupyterLab, Vim, L^AT_EX, Markdown**Systems:** Unix/Linux, macOS, Windows

WORK EXPERIENCE

University of California, Berkeley

Berkeley, CA

*Graduate Student Researcher, Whaley Group**Sep 2018 – Aug 2023*

- Led an interdisciplinary initiative spanning quantum computing and computational chemistry, coordinating multiple research teams to design variational quantum-classical algorithms for electronic structure simulation on near-term devices, achieving chemical-accuracy benchmarks.
- Rewrote critical parts of the Python research codebase enabling the new algorithm family; mentored graduate students who extended the framework in subsequent research.
- Engineered resource-efficient quantum circuit architectures using tensor network methods to optimize variational quantum states and reduce circuit cost.
- Applied rigorous numerical linear algebra and large-scale optimization techniques across projects, translating theoretical models into scalable implementations.

University of California, Berkeley

Berkeley, CA

*Graduate Student Researcher, Quantum Nanoelectronics Lab**Sep 2018 – May 2019*

- Built FPGA-based hardware (QubiC) for real-time control and readout of superconducting qubits in collaboration with LBNL scientists and engineers; work culminated in an APS March Meeting talk on low-latency quantum feedback.
- Installed and relocated dilution refrigerators and cryogenic hardware, including transferring John Clarke’s refrigerator to the current lab and assembling 80/20 framing as part of the infrastructure setup.
- Developed cross-stack understanding of quantum systems by working at the interface of theory, experiment, and engineering teams.

University of California, Berkeley

Berkeley, CA

*Graduate Student Instructor**Fall 2017 – Spring 2018, Spring 2020*

- **Head Teaching Assistant, Quantum Information Science & Technology (Physics C191):** Led weekly discussion sections and prepared exams, problem sets, and instructional materials.
- **Awards:** Outstanding Graduate Student Instructor Award, for excellence in teaching.

PUBLICATIONS & SELECT PRESENTATIONS

- **Unpil Baek**, Diptarka Hait, James Shee, Oskar Leimkuhler, William J. Huggins, Torin F. Stetina, Martin Head-Gordon, and K. Birgitta Whaley. “Say NO to Optimization: A Nonorthogonal Quantum Eigensolver.” *PRX Quantum*, 2023.
- William J. Huggins, Joonho Lee, **Unpil Baek**, Bryan O’Gorman, and K. Birgitta Whaley. “A Non-Orthogonal Variational Quantum Eigensolver.” *New Journal of Physics*, 2020.
- **Unpil Baek**, Yilun Xu, Gang Huang, Lawrence Doolittle, and Irfan Siddiqi. “An FPGA-based quantum feedback system for real-time qubit control.” *APS March Meeting*, Boston, MA, USA, March 7, 2019.

HONORS, AWARDS & DISTINCTIONS

- **Full Membership, Sigma Xi**: Sep 2025-present
- **Outstanding Graduate Student Instructor Award**: UC Berkeley, May 2018
- **Departmental Citation in Physics**: UC Berkeley, May 2017
- **Dorothea Klumpke Roberts Prize in Mathematics**: UC Berkeley, May 2017
- **Junior Initiate, Phi Beta Kappa (PBK)**: Mar 2016-present
- **Undergraduate Dean’s Honors List**: UC Berkeley, 2013–2017

OTHER

Languages: *Fluent*: English, Korean. *Conversational*: Japanese, Mandarin Chinese

Interests: Jazz piano (performer & educator), world music, foreign languages, reading, travel